HS-OBS	Functional Requirements Specification
DEPARTMENT	DOCUMENT TYPE

VZI 6

CCTV SFRS

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Revision History

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01	27/01/2020	R. Luppino	“EC01P009001B_CCTV_SFRS_R01” for previous fleet – DOORS BL 1.3
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1. Introduction

~~This document is a function specification for the control of the CCTV system for V300ZEFIRO for customer Trenitalia.~~

This document is a functional requirements specification for the control of the *VZI6 CCTV system* for Trenitalia customer.

The new VZI6 units incorporates some other contractual requests concerning the change in layout requested by the customer.

The new fleet also implements the TRDP protocol for communications.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:

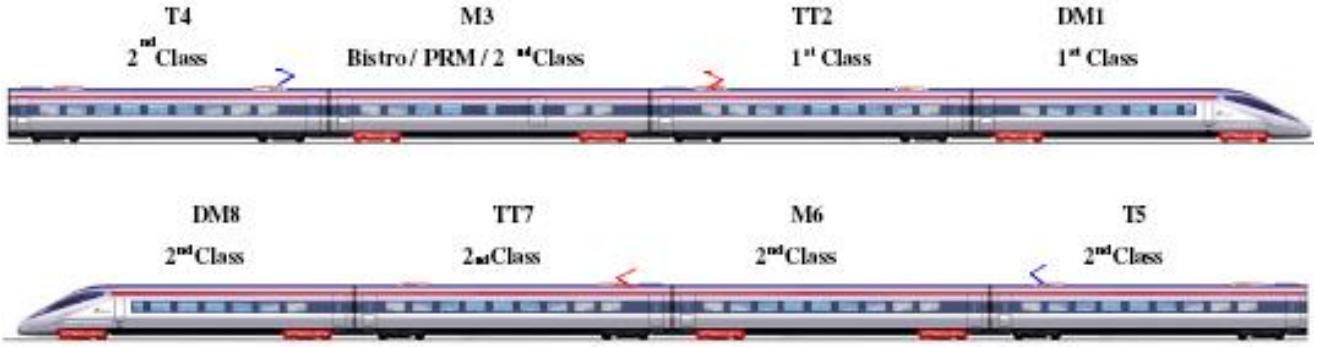


Figure 1 – VZI 6 Train Configuration

- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

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1.2 Definitions and abbreviations

CD	Circuit Diagram
CCTV	Closed Circuit Tele Vision
DVR	Digital Video Recorder
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
LCU	Local Control Unit
MAR	Mobile Access Router
MMI	Man Machine Interface
PIS	Passenger Information System
SCU	System Control Unit
TCMS	Train Control and Management System

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2. System Overview

The video surveillance system is completely integrated into the PIS system **[RD 1, RD 2 and RD 5**].

There will be one camera for each vestibule and two cameras for each passenger compartment for end cars and standard cars.

The M3 car will have one camera for the passenger compartment, one for each vestibule and one for the bar bistro area.

Therefore, there will be four cameras for the intermediate cars while there will be three cameras for the end cars.

The video streams are recorded to guarantee 100 hours of recording. The download of information can be anyway performed from a centralized location though the PIS network.

The option for the admission on the Swiss railway network will include two external cameras for each cab to allow the rear view mirror function for each side of the train. In this case two additional LCD monitors will be installed inside the walls of each cab, one for each side to be monitored.

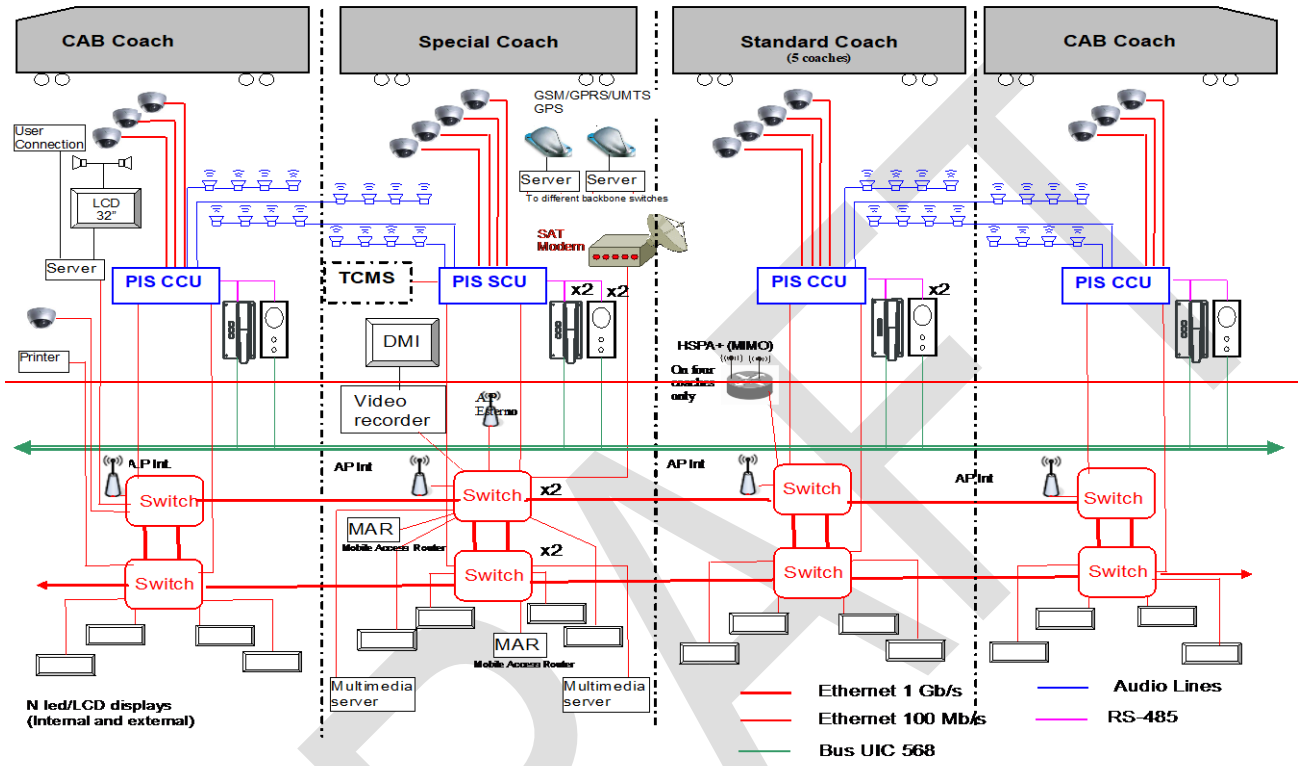
Cameras and monitors can be activated automatically with train at zero speed or can be activated manually from drivers display.

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2.1 System Configuration

The CCTV system is integrated together with PIS system.

The following figure is a scheme of the system architecture.



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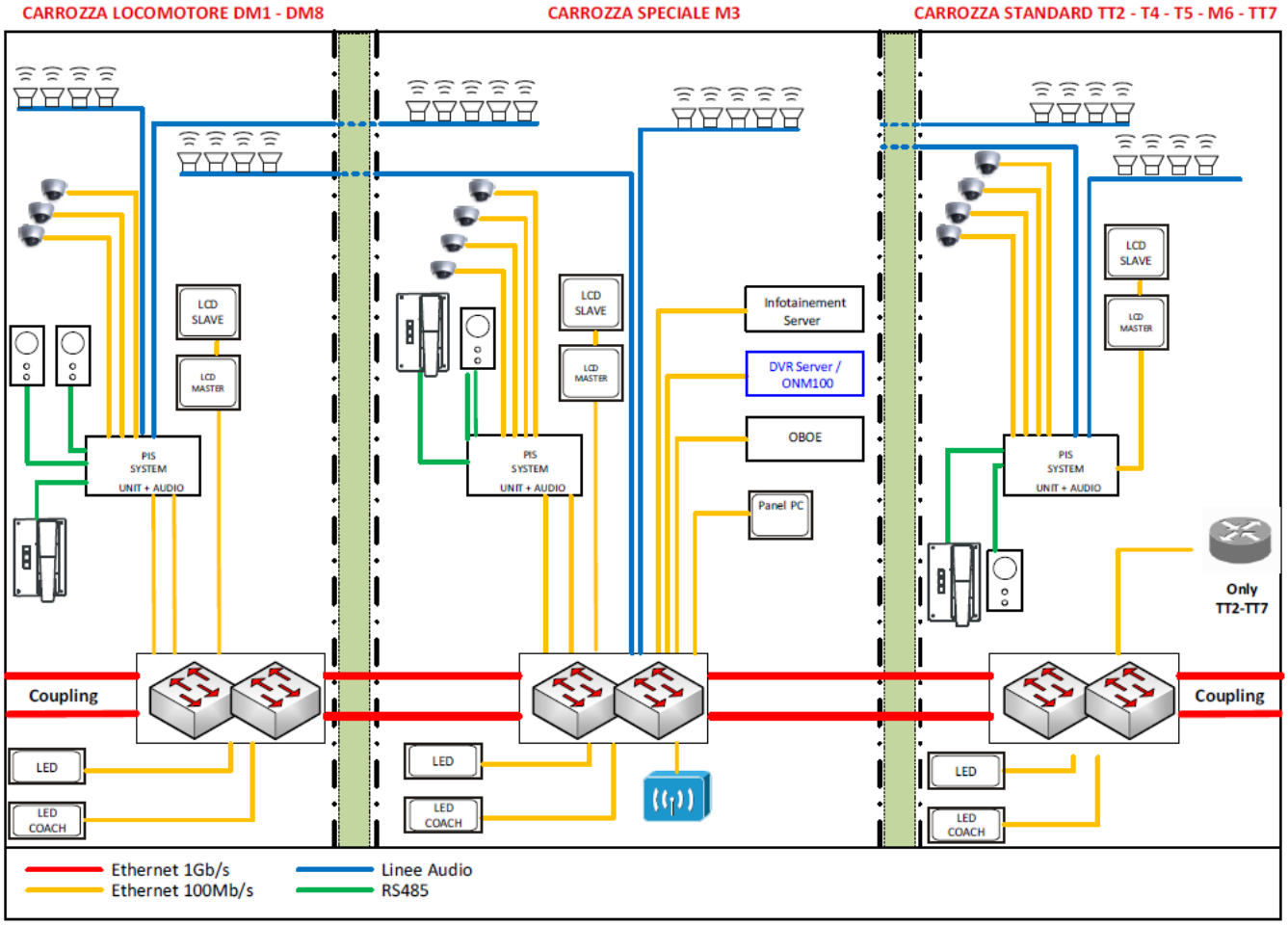


Figure 2 - PIS and CCTV System Architecture

The cameras distribution in the train is according to the following figure:

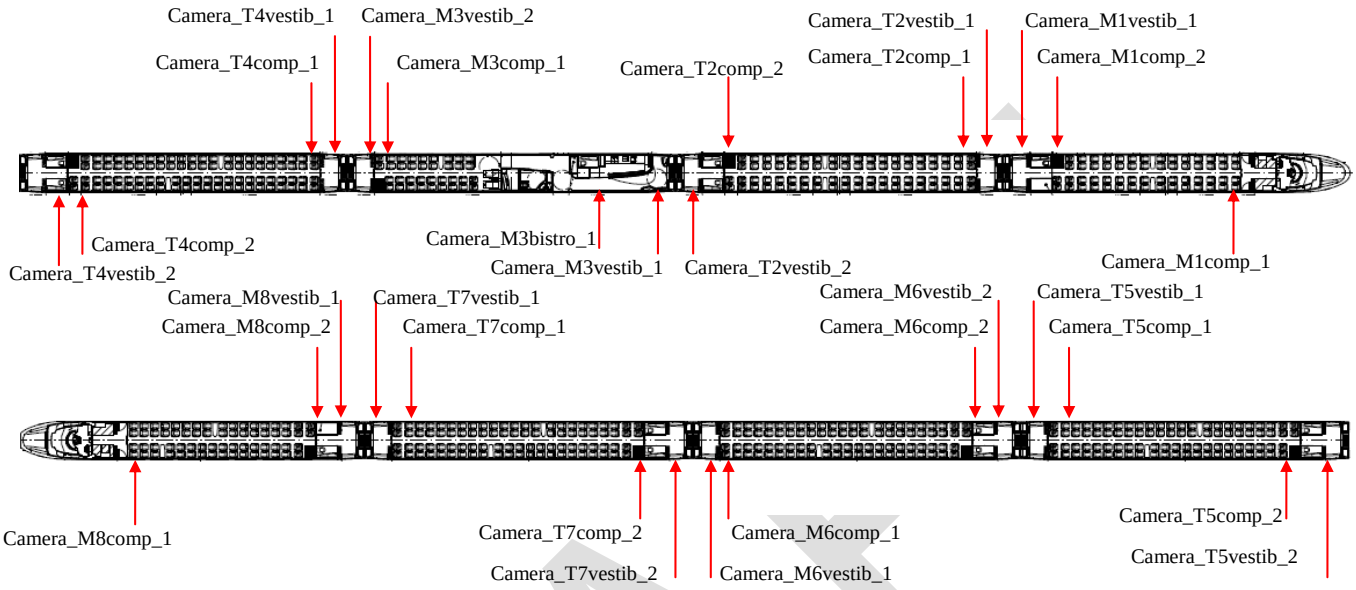


Figure 3 - CCTV Cameras Layout

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3. Functional requirements

3.1 General CCTV Control

3.1.1 Architecture

3.1.1.1 Function design

The CCTV is completely integrated into the PIS system. There is no TCMS interaction for the video recording functions. All the information is available in the PIS system.

Information that have to be recorded together with images (Train Number, Date Time, and vehicle number) are already available to the PIS/CCTV system. The train number is received ~~through GSM~~ by the ground infrastructure (OBoE functionality). The Date Time is computed from GPS signal. Vehicle number is known from PIS system [**RD 1**].

In case of degraded situation (loss of ground communication) the train number can be set manually through the PIS MMI.

3.1.1.1.1 Normal condition

Train number and date time information is received by the OBoE ~~through GSM and~~ and sent to the DVR (Digital Video Recorder) for image tagging.

3.1.1.1.2 Failure cases

A. failure of GPS => time recorded with images could be not correct
B. failure of GSM connection => train number to be registered with images could be not available
C. failure of PIS network => train number and data time to be registered with images could be not available

3.1.1.1.3 Failure detection

A. Failure detected by the PIS and communicated to TCMS
B. Failure detected by the PIS and communicated to TCMS
C. Failure detected by the PIS and communicated to TCMS

3.1.1.1.4 Failure handling

A. Failure isolated and repaired during maintenance.
B. PIS shall warn the train staff that can manually insert train number. Failure isolated and repaired during maintenance.
C. Failure isolated and repaired during maintenance.

3.1.1.2 Requirements

ID	2F_13.Zefiro-Europe.TRS.40
Description	Each image shall be recorded together with the information such as: * Train number * Date and time * Vehicle number
Safety level	N/A

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Nr	Description	Type	Derived to	User Interface	SIL	China FVDS
2F_13.Zefiro-Europe.CONCEPT.55	CCTV shall use Train Number, Date Time, and veihcle number when storing CCTV images	Derived Requirement	4S_09.05.-.Video Surveillance Systems and Closed Circuit Televisions		N/A	

3.1.1.3 Option for the admission on the Swiss railway network

3.1.1.3.1 Function design

The cab will be equipped with two cameras and two displays (on each side of the cab) to act as rear-view mirror

3.1.1.3.1.1 Normal condition

When the train is at standstill the two display will show the images of the two cameras. As soon as the train starts the display are switched off

3.1.1.3.1.2 Failure cases

A. TCMS fails to command the display to be switched off => display on while the train is driving.

3.1.1.3.1.3 Failure detection

A. Failure detected by the driver.

3.1.1.3.1.4 Failure handling

A. Failure isolated and repaired during maintenance.

3.1.1.3.2 Requirements

ID	2F_13.Zefiro-Europe.TRS.23
Description	Deleted
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_13.Zefiro-Europe.CONCEPT.60	The TCMS shall command the PIS to switch on the displays in the active cab when the train speed is 0 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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Nr	Description	Type	Derived to	User Interface	SIL
2F_13.Zefiro-Europe.CONCEPT.101	The PIS shall turn on the CCTV displays in the active can when commanded by the TCMS.	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A

3.1.2 PIS MMI in staff area

3.1.2.1 Function design

The camera images are normally displayed on the PIS MMI. In case of fire alarm detected on-board the Staff MMI shall visualize the image of the nearest camera to the location of where the fire has been detected.
In case of emergency intercom activation, the MMI shall visualize the image of the camera nearer to the Intercom. This function is completely realized by the PIS/CCTV system.

3.1.2.1.1 Normal condition

When fire is detected on-board the Staff MMI visualizes the image of the nearest camera to the location of where the fire has been detected.

3.1.2.1.2 Failure cases

A. TCMS fails to command PIS MMI to display the camera near the fire alarm.

3.1.2.1.3 Failure detection

A. Failure not detected.

3.1.2.1.4 Failure handling

A. Failure not detected.

3.1.2.2 Requirements

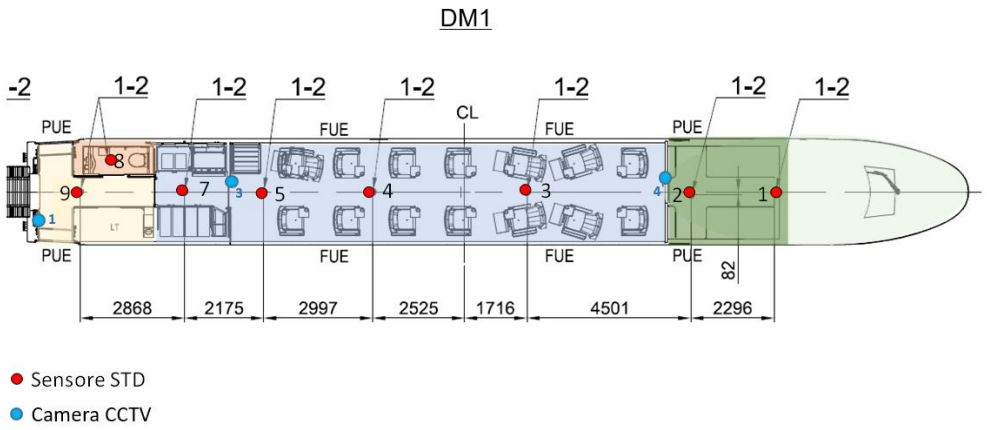
Refer to **RD 6** for smoke sensors details.

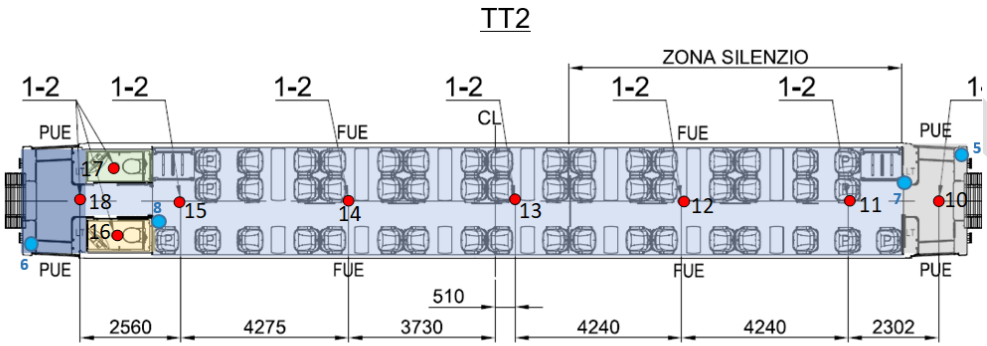
ID	2F_13.Zefiro-Europe.TRS.37
Description	In case of fire alarm detected onboard the Staff MMI shall visualize the image of the nearest camera to the location of where the fire has been detected.
Safety level	N/A

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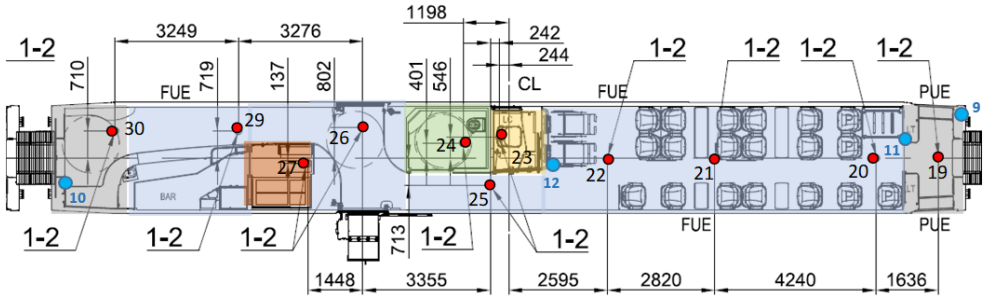
Nr	Description	Type	Derived to	User Interface	SIL
2F_13.Zefiro-Europe.CONCEPT.65	In case of fire alarm the TCMS shall send to PIS SCU indication of the camera nearest to the detection of fire, according below concepts. In case of multiple alarm, TCMS shall send the indication of camera in sequence.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

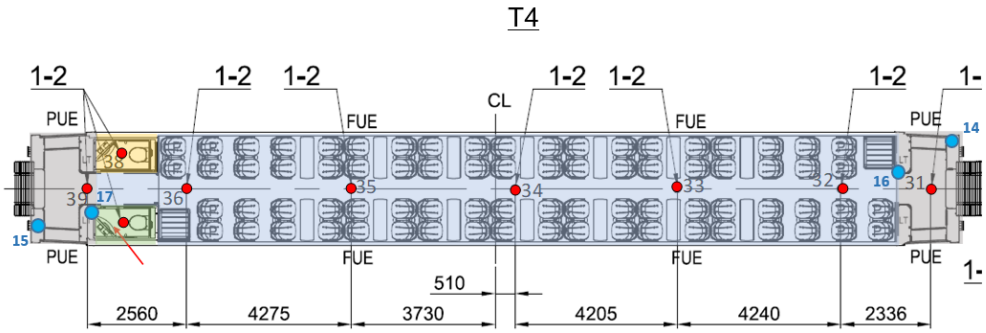
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~~2F_13.Zefiro-Europe.CONCEPT.146~~ 2F_0y.0z.Zefiro-Europe.CONCEPT.HR146	When a fire alarm is set in DM1 car, a cam shall be activated according following relation:  ● Sensore STD ● Camera CCTV **item** / **smoke sensor ID n°** / **installed in:** / **Camera ID** 1 / SD01 / smoke sensor driver cab / No camera 2 / SD02 / smoke sensor electronic room / No camera 3 / SD03 / smoke sensor saloon 1 / 3 4 / SD04 / smoke sensor saloon 2 / 4 5 / SD05 / smoke sensor corridor / 4 6 / SD06 / smoke sensor meeting room / 4 7 / SD07 / smoke sensor galley / No camera 8 / SD08 / smoke sensor toilet (executive) / 1 9 / SD09 / smoke sensor vestibule / 1	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

~~2F_13.Zefiro-Europe.CONCEPT.147~~ 2F_0y.0z.Zefiro-Europe.CONCEPT.HR147	When a fire alarm is set in TT2 car, a cam shall be activated according following relation:  ● Sensore STD ● Camera CCTV **item** / **smoke sensor ID n°** / **installed in:** / **Camera ID** 10 SD10 smoke sensor short vestibule 5 11 SD11 smoke sensor saloon 1 7 12 SD12 smoke sensor saloon 2 7 13 SD13 smoke sensor saloon 3 7 14 SD14 smoke sensor saloon 4 8 15 SD15 smoke sensor saloon 5 8 16 SD16 smoke sensor toilet 1 (left) 6 17 SD17 smoke sensor toilet 2 (right) 6 18 SD18 smoke sensor vestibule 6	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

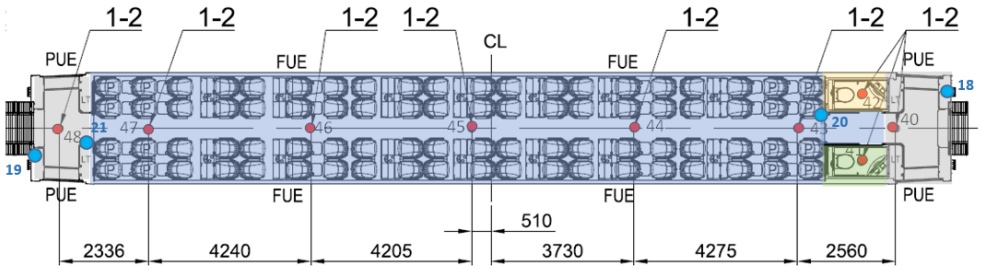
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~~2F_13.Zefiro-Europe.CONCEPT.148~~ 2F_0y.0z.Zefiro-Europe.CONCEPT.HR148	When a fire alarm is set in M3 car, a cam shall be activated according following relation: M3  ● Sensore STD ● Camera CCTV **item** / **smoke sensor ID n°** / **installed in:** / **Camera ID** 19 / SD19 / smoke sensor short vestibule / 9 20 / SD20 / smoke sensor saloon 1 / 12 21 / SD21 / smoke sensor saloon 2 / 11 22 / SD22 / smoke sensor saloon 3 / 11 23 / SD23 / smoke sensor train staff / No camera 24 / SD24 / smoke sensor toilet HK / No camera 25 / SD25 / smoke sensor corridor 1 / No camera 26 / SD26 / smoke sensor corridor 2 / No camera 27 / SD27 / smoke sensor crew room / No camera 28 / SD28 / smoke sensor crew toilet / No camera 29 / SD29 / smoke sensor bistrot 1 / 10 30 / SD30 / smoke sensor bistrot 2 / 10	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

~~2F_13.Zefiro-Europe.CONCEPT.149~~ 2F_0y.0z.Zefiro-Europe.CONCEPT.HR149	When a fire alarm is set in T4 car, a cam shall be activated according following relation:  ● Sensore STD ● Camera CCTV **item** / **smoke sensor ID n°** / **installed in:** / **Camera ID** 31 / SD31 / smoke sensor short vestibule / 14 32 / SD32 / smoke sensor saloon 1 / 16 33 / SD33 / smoke sensor saloon 2 / 16 34 / SD34 / smoke sensor saloon 3 / 16 35 / SD35 / smoke sensor saloon 4 / 17 36 / SD36 / smoke sensor saloon 5 / 17 37 / SD37 / smoke sensor toilet 1 (left) / 15 38 / SD38 / smoke sensor toilet 2 (right) / 15 39 / SD39 / smoke sensor vestibule / 15	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

~~2F_13.Zefiro-Europe.CONCEPT.150~~ 2F_0y.0z.Zefiro-Europe.CONCEPT.HR150	When a fire alarm is set in T5 car, a cam shall be activated according following relation:	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

T5

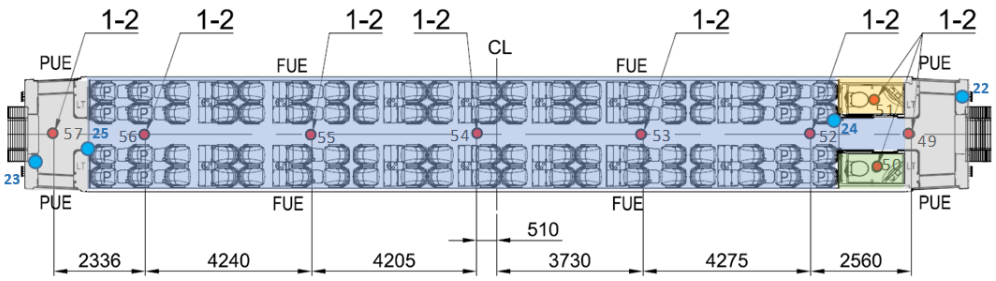


● Sensore STD
● Camera CCTV

item	smoke sensor ID n°	installed in:	Camera ID
40	SD40	smoke sensor vestibule	18
41	SD41	smoke sensor toilet 1 (left)	18
42	SD42	smoke sensor toilet 2 (right)	18
43	SD43	smoke sensor saloon 5	20
44	SD44	smoke sensor saloon 4	20
45	SD45	smoke sensor saloon 3	20
46	SD46	smoke sensor saloon 2	21
47	SD47	smoke sensor saloon 1	21
48	SD48	smoke sensor short vestibule	19

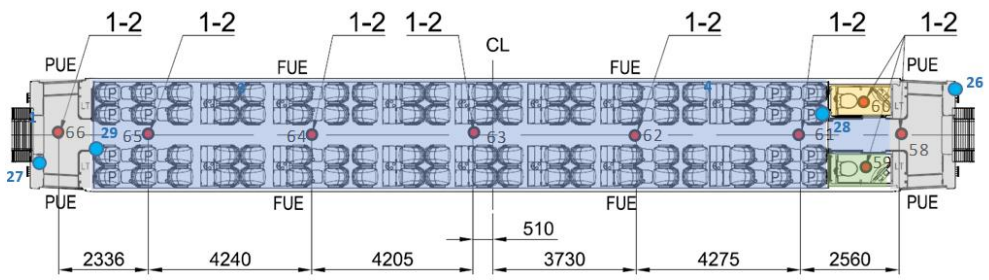
~~2F_13.Zefiro-Europe.CONCEPT.151~~ 2F_0y.0z.Zefiro-Europe.CONCEPT.HR151	When a fire alarm is set in M6 car, a cam shall be activated according following relation:	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

M6

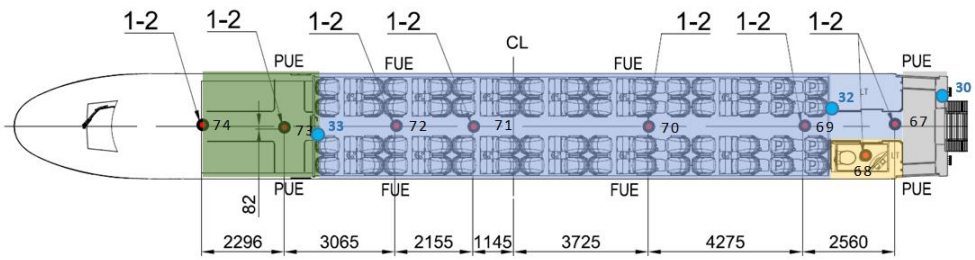


● Sensore STD
● Camera CCTV

item	smoke sensor ID n°	installed in:	Camera ID
49	SD49	smoke sensor vestibule	22
50	SD50	smoke sensor toilet 1 (left)	22
51	SD51	smoke sensor toilet 2 (right)	22
52	SD52	smoke sensor saloon 5	24
53	SD53	smoke sensor saloon 4	24
54	SD54	smoke sensor saloon 3	24
55	SD55	smoke sensor saloon 2	25
56	SD56	smoke sensor saloon 1	25
57	SD57	smoke sensor short vestibule	23

~~2F_13.Zefiro-Europe.CONCEPT.152~~ 2F_0y.0z.Zefiro-Europe.CONCEPT.HR152	When a fire alarm is set in TT7 car, a cam shall be activated according following relation: TT7  Sensore STD Camera CCTV	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

item	smoke sensor ID n°	installed in:	Camera ID
58	SD58	smoke sensor vestibule	26
59	SD59	smoke sensor toilet 1 (left)	26
60	SD60	smoke sensor toilet 2 (right)	26
61	SD61	smoke sensor saloon 5	28
62	SD62	smoke sensor saloon 4	28
63	SD63	smoke sensor saloon 3	28
64	SD64	smoke sensor saloon 2	29
65	SD65	smoke sensor saloon 1	29
66	SD66	smoke sensor short vestibule	27

~~2F_13.Zefiro-Europe.CONCEPT.153~~ 2F_0y.0z.Zefiro-Europe.CONCEPT.HR153	When a fire alarm is set in DM8 car, a cam shall be activated according following relation: DM8  ● Sensore STD ● Camera CCTV **item** / **smoke sensor ID n°** / **installed in:** / **Camera ID** 67 / SD67 / smoke sensor vestibule / 30 68 / SD68 / smoke sensor toilet / 30 69 / SD69 / smoke sensor saloon 4 / 32 70 / SD70 / smoke sensor saloon 3 / 32 71 / SD71 / smoke sensor saloon 2 / 33 72 / SD72 / smoke sensor saloon 1 / 33 73 / SD73 / smoke sensor electronic room / No camera 74 / SD74 / smoke sensor driver cab / No camera	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_13.Zefiro-Europe.CONCEPT.108	The PIS MMI in the train staff compartment shall queue the indication of the TCMS and visualize the image of the camera indicated by the TCMS	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A
2F_13.Zefiro-Europe.CONCEPT.107	In case of activation of the passenger emergency intercom the MMI shall visualize the image of the nearest camera to the location of the relevant intercom device.	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A

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3.2 Events

3.2.1 Function Design

All the system diagnostic shall be sent to the TCMS thorough Ethernet.

Refer to **RD 3 and RD 4** for the complete list signal.

3.2.2 Requirements

ID	2F_11.Zefiro-Europe.TRS.224
Description	Diagnostics from the supplier shall be stored in the train diagnostic system.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_13.Zefiro-Europe.CONCEPT.76	Diagnostics from the supplier shall be stored in the train diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

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4. Signal Interface

~~1.1 I/O Interface~~

~~1.1.1 Supplier system~~

~~Inputs:~~

Nr	Functional name	Type	Linked to requirement
~~2F_13.Zefiro-Europe.CONCEPT.28~~			

~~Outputs:~~

Nr	Functional name	Type	Linked to requirement
~~2F_13.Zefiro-Europe.CONCEPT.29~~			

~~1.1.2 TCMS SW~~

~~Inputs:~~

Nr	Functional name	Type	Linked to requirement
~~2F_13.Zefiro-Europe.CONCEPT.30~~			

~~Outputs:~~

Nr	Functional name	Type	Linked to requirement
~~2F_13.Zefiro-Europe.CONCEPT.31~~			

4.1 Bus Interface

Supplier to TCMS SW:

TCMS SW to Supplier:

See also annex for cameras-to-signals links.

Nr	Functional name	Type	Linked to requirement	Signal [Code]
2F_13.Zefiro-Europe.CONCEPT.32	RearViewDmiON	L	2F_13.Zefiro-Europe.CONCEPT.60	CRearViewDmiOn
2F_13.Zefiro-Europe.CONCEPT.110	EnabledCab	L	2F_13.Zefiro-Europe.CONCEPT.60	IEnabledCab
2F_13.Zefiro-Europe.CONCEPT.33	Camera_M1vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 01]
2F_13.Zefiro-Europe.CONCEPT.113	Camera_M1comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 03]
2F_13.Zefiro-Europe.CONCEPT.114	Camera_M1comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 04]
2F_13.Zefiro-Europe.CONCEPT.115	Camera_T2vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 05]

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Nr	Functional name	Type	Linked to requirement	Signal [Code]
2F_13.Zefiro-Europe.CONCEPT.118	Camera_T2vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 06]
2F_13.Zefiro-Europe.CONCEPT.116	Camera_T2comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 07]
2F_13.Zefiro-Europe.CONCEPT.117	Camera_T2comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 08]
2F_13.Zefiro-Europe.CONCEPT.119	Camera_M3vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 09]
2F_13.Zefiro-Europe.CONCEPT.120	Camera_M3vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 10]
2F_13.Zefiro-Europe.CONCEPT.121	Camera_M3comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 11]
2F_13.Zefiro-Europe.CONCEPT.122	Camera_M3comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 12]
2F_13.Zefiro-Europe.CONCEPT.123	Camera_T4vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 14]
2F_13.Zefiro-Europe.CONCEPT.124	Camera_T4vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 15]
2F_13.Zefiro-Europe.CONCEPT.125	Camera_T4comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 16]
2F_13.Zefiro-Europe.CONCEPT.126	Camera_T4comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 17]
2F_13.Zefiro-Europe.CONCEPT.127	Camera_T5vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 18]
2F_13.Zefiro-Europe.CONCEPT.128	Camera_T5vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 19]
2F_13.Zefiro-Europe.CONCEPT.129	Camera_T5comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 20]
2F_13.Zefiro-Europe.CONCEPT.130	Camera_T5comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 21]
2F_13.Zefiro-Europe.CONCEPT.131	Camera_M6vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 22]
2F_13.Zefiro-Europe.CONCEPT.132	Camera_M6vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 23]
2F_13.Zefiro-Europe.CONCEPT.133	Camera_M6comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 24]
2F_13.Zefiro-Europe.CONCEPT.134	Camera_M6comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 25]
2F_13.Zefiro-Europe.CONCEPT.135	Camera_T7vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 26]

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Nr	Functional name	Type	Linked to requirement	Signal [Code]
2F_13.Zefiro-Europe.CONCEPT.136	Camera_T7vestib_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 27]
2F_13.Zefiro-Europe.CONCEPT.137	Camera_T7comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 28]
2F_13.Zefiro-Europe.CONCEPT.138	Camera_T7comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 29]
2F_13.Zefiro-Europe.CONCEPT.139	Camera_M8vestib_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 30]
2F_13.Zefiro-Europe.CONCEPT.140	Camera_M81comp_1	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 32]
2F_13.Zefiro-Europe.CONCEPT.141	Camera_M8comp_2	L	2F_13.Zefiro-Europe.CONCEPT.65	CCctvActiveCamReq [code = 33]

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5. References

RD	Code	Title	Source	Revision
RD 1	S23025.03.1015TR	VZI-6 PIS - Functional Description	Leonardo	01.00
RD 2	S23025.03.1028TR	Functional Diagram – PIS PA TVCC	Leonardo	01.00
RD 3	EC130021261	Interface Control Document Requirements for PIS	Hitachi	00.03
RD 4	EC130021262	ICD Requirements for PIS Annex	Hitachi	00.03
RD 5	FA020025354	Passenger Information and Entertainment – VZI 6 SFRS	Hitachi	00
RD 6	FA020025419	SFRS - Fire Protection System	Hitachi	AA

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6. Annex: Relations between smoke sensors and cameras

Signal Code	Description	Size	Coding
CCctvActiveCamReq	Active CCTV camera request	UINT8	~~0 = No cam selected~~ ~~1 = Cam DM1 Vestibule 1~~ ~~2 = spare~~ ~~3 = Cam DM1 Comp 1~~ ~~4 = Cam DM1 Comp 2~~ ~~5 = Cam TT2 Vestibule 1~~ ~~6 = Cam TT2 Vestibule 2~~ ~~7 = Cam TT2 Comp 1~~ ~~8 = Cam TT2 Comp 2~~ ~~9 = Cam M3 Vestibule 1~~ ~~10 = Cam M3 Vestibule 2~~ ~~11 = Cam M3 Comp 1~~ ~~12 = Cam M3 Comp 2~~ ~~13 = spare~~ ~~14 = Cam T4 Vestibule 1~~ ~~15 = Cam T4 Vestibule 2~~ ~~16 = Cam T4 Comp 1~~ ~~17 = Cam T4 Comp 2~~ ~~18 = Cam T5 Vestibule 1~~ ~~19 = Cam T5 Vestibule 2~~ ~~20 = Cam T5 Comp 1~~ ~~21 = Cam T5 Comp 2~~ ~~22 = Cam M6 Vestibule 1~~ ~~23 = Cam M6 Vestibule 2~~ ~~24 = Cam M6 Comp 1~~ ~~25 = Cam M6 Comp 2~~ ~~26 = Cam TT7 Vestibule 1~~ ~~27 = Cam TT7 Vestibule 2~~ ~~28 = Cam TT7 Comp 1~~ ~~29 = Cam TT7 Comp 2~~ ~~30 = Cam DM8 Vestibule 1~~ ~~31 = spare~~ ~~32 = Cam DM8 Comp 1~~ ~~33 = Cam DM8 Comp 2~~ All available codes in ICD: 0 = No cam selected 1 = Cam DM1 Vestibule 1 2 = Cam DM1 Vestibule 2 3 = Cam DM1 Comp 1

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			4 = Cam DM1 Comp 2 5 = Cam TT2 Vestibule 1 6 = Cam TT2 Vestibule 2 7 = Cam TT2 Comp 1 8 = Cam TT2 Comp 2 9 = Cam M3 Vestibule 1 10 = Cam M3 Vestibule 2 11 = Cam M3 Comp 1 12 = Cam M3 Comp 2 13 = Cam M3 Bistro 1 14 = Cam T4 Vestibule 1 15 = Cam T4 Vestibule 2 16 = Cam T4 Comp 1 17 = Cam T4 Comp 2 18 = Cam T5 Vestibule 1 19 = Cam T5 Vestibule 2 20 = Cam T5 Comp 1 21 = Cam T5 Comp 2 22 = Cam M6 Vestibule 1 23 = Cam M6 Vestibule 2 24 = Cam M6 Comp 1 25 = Cam M6 Comp 2 26 = Cam TT7 Vestibule 1 27 = Cam TT7 Vestibule 2 28 = Cam TT7 Comp 1 29 = Cam TT7 Comp 2 30 = Cam DM8 Vestibule 1 31 = Cam DM8 Vestibule 2 32 = Cam DM8 Comp 1 33 = Cam DM8 Comp 2

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